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MILITARY SPECIFICATION

# MECHANISMS, RECOIL, M3A3 AND M4A3 (W/REPAIR PARTS AND W/O EQUIPMENT OR ACCESSORIES)

*This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.*

## 1. SCOPE

1.1 This specification covers the requirements for one type of hydro-pneumatic recoil mechanism known as the "Mechanism, Recoil M3A3" model when assembled with the 155 MM Gun Carriage M1, and as the "Mechanism, Recoil, M4A3" model when assembled with 8-inch Howitzer Carriage M1.

## 2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification:

### SPECIFICATIONS

MILITARY

MIL-P-116 — Preservation, Methods of.

MIL-M-11965 — Packaging of Recoil Mechanisms M21, M23, M3, M4.

MIL-W-14539 — Weapon Parts for Anti-aircraft Artillery; Rocket Launchers and Small Arms, Packaging of.

LSQAP 7148356 — Indexes of Supplementary Quality

LSQAP 7148374 — Assurance Provisions.

### STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-173 — Painting and Finishing Systems for Artillery Material.

### DRAWINGS

ORDNANCE CORPS

F7148356 — Recoil Mechanism M3A3.

F7148374 — Recoil Mechanism M4A3.

GLB114583 — Index to List of Inspection Gages and Parts Gaged on the Mechanisms Recoil M3A3 and M4A3.

**(Copies of applicable documents required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)**

## 3. REQUIREMENTS

**3.1 Dimensions.** Dimensions of parts and assemblies shall conform to the Ordnance drawings **F7148356** and **F7148374.**

**3.2 Performance and product characteristics.**

FSC 1025
FSC 1030

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3.2.1 *Proof-firing.* Completed mechanisms shall be fully assembled and lubricated for proof-firing. Each mechanism shall function as a coordinated assembly without evidence of malfunction. Material failures, defects of manufacture or assembly, or other faults attributable to the mechanism shall be the responsibility of the contractor.

3.2.2 *Gymnastication.* After final assembly each recoil mechanism shall be operated mechanically in recoil and counter recoil for 75 to 100 cycles. The recoil mechanism shall operate without erratic movement or malfunctioning and shall show no indication of excessive leakage (see 4.2.4.2).

3.2.3 *Packings.* Packings shall be carefully adjusted so that the mechanism will function with a minimum of friction (table I) and with only sufficient leakage to provide necessary lubrication. The friction of the floating piston and recoil rod and counter recoil piston rod shall be within the limits specified in table I after assembly, after gymnastication, and after proof-firing.

TABLE I. *Friction in pounds per square inch of hydraulic pressure after proof-firing*

| Part Name | Minimum* | Maximum* |
|---|---|---|
| Floating Piston | 70 | 350 |
| Recoil Piston Rod and Counter Recoil Piston Rod | 110 | 350 |

* Buffer must be five inches out of battery when friction values are taken.

3.2.4 *Recoil system.* The recoil system shall be free from voids or trapped air which will affect the functioning of the mechanism or cause the oil to appear foamy or emulsified.

3.2.5 *Oil index.* The oil index shall function without evidence of interference throughout its entire range of travel. There shall be only sufficient leakage to provide necessary lubrication.

3.2.6 *Air filling valve.* The air filling valve shall show no indication of leakage.

3.3 **Painting.** Unless otherwise specified the mechanisms shall be painted in accordance with Standard MIL–STD–173.

## 4. QUALITY ASSURANCE PROVISIONS

### 4.1 General quality assurance provisions.

4.1.1 *Contractor inspection.* Unless otherwise specified herein, the inspections required by this specification shall be performed by the contractor.

4.1.2 *Contractor quality assurance system.* The contractor shall provide and maintain an effective quality assurance system acceptable to the Government covering the supplies under this contract. Results of all examinations and tests performed under the quality assurance system shall be made available to the Government. A current written description will be considered acceptable when, as a minimum, it provides the quality assurance required by the detail specification and other documents referenced in the detail specification. The contractor shall not be restricted to the inspection station or to the method of inspection listed provided that an equivalent control is included in the approved quality assurance procedure. In cases of dispute, as to whether certain procedures of the system provide equal assurance, the comparable procedure of the detail specification shall be used. The contractor shall notify the Government of, and obtain approval for, any change to the written procedure that affects the degree of assurance required by the detail specification or other documents referenced therein.

4.1.3 *Government verification.* All quality assurance operations performed by the contractor shall be subject to Government verification at unscheduled intervals. Verification will consist of surveillance of the operations to determine whether the practices, methods and procedures of the written inspection plan are being properly applied, and Government product inspection to measure quality of product offered for acceptance. Deviation from the prescribed or agreed upon procedures, or in-



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stances of poor practices which might have an effect upon the quality of the product shall be immediately called to the attention of the contractor. Failure of the contractor to promptly correct deficiencies discovered shall be cause for suspension of acceptance until correction has been made or until conformance of product to prescribed criteria has been demonstrated. To avoid interference with operations, the contractor shall designate a responsible official or officials to whom the Government inspector will report such instances.

**4.2 Inspection provisions.**

4.2.1 *Lot formation.* Lots shall be as large as practical and shall consist of products made under essentially the same conditions and from the same batch of material.

4.2.2 *Sampling.*

4.2.2.1 *Assembled mechanisms.* Unless otherwise specified, each assembled mechanism shall be inspected for compliance with this specification.

4.2.2.2 *Parts and assemblies.* Sampling of parts and assemblies for inspection shall be as specified in Standard MIL–STD–105.

4.2.2.3 *Packaging.* Sampling of packaging for inspection shall be as specified in Specification MIL–P–116.

4.2.3 *Examination.*

4.2.3.1 *Material examination.* The contractor shall provide the Government inspector with a certified statement that the material(s) used in all parts and assemblies of the mechanism conform(s) to the material specification(s) referenced on the applicable drawing(s).

4.2.3.2 *Assembled mechanisms.* Each assembled mechanism shall be examined visually for presence and continuity of protective coating and for the correctness and legibility of serial numbers (see 6.2).

4.2.3.3 *Parts and assemblies.* Examination of parts and assemblies shall be as specified in the applicable Supplementary Quality Assurance Provision (SQAP) forming a part of this specification.

4.2.3.4 *Packaging.* Examination of packaging shall be as specified in Specification MIL–P–116.

4.2.4 *Inspection testing.*

4.2.4.1 *Proof-firing test.* The mechanism shall perform in accordance with 3.2.1 without indication of malfunction. Failure of the mechanism to perform as intended shall be cause for rejection. Proof-firing shall consist of firing standard rounds as defined in 4.2.4.1.1. Proof-firing test shall be conducted in accordance with 4.3.1. Proof-firing will be performed by the Government.

4.2.4.1.1 *Definitions.* The terms employed in proof-firing requirements are defined as follows:

(a) For the Recoil Mechanism M3A3—
The standard weight projectile is 95 pounds.
A standard round is one employing a standard weight projectile and loaded to give a muzzle velocity of 2,800 feet per second.
The rated maximum pressure is 40,000 pounds per square inch.
Powder used in the rounds prescribed in table II shall be from a lot that gives the standard weight projectile a muzzle velocity of 2,800 feet per second with an average maximum powder pressure of not less than 37,000 pounds per square inch nor more than 40,000 pounds per square inch.

(b) For the Recoil Mechanism M4A3—
The standard weight projectile is 200 pounds.
A standard round is one employing a standard weight projectile and



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loaded to give a muzzle velocity of 1,950 feet per second.

The rated maximum pressure is 33,000 pounds per square inch.

Powder used in the rounds prescribed in table II shall be from a lot that gives the standard weight projectile a muzzle velocity of 1,950 feet per second with an average maximum powder pressure of not less than 30,000 pounds per square inch nor more than 33,000 pounds per square inch.

4.2.4.2 *Packings.* After gymnastication, packings shall be examined visually to determine that the amount of oil escaping during gymnastication is not in excess of that required for lubrication. Excess oil escape can be determined by evidence of oil drip.

4.2.4.3 *Alinement.* The mechanism shall be checked for alinement in accordance with the applicable drawings. The contractor shall ascertain that the axis of the recoil cylinder, piston rods, and piston rod stuffing boxes are in alinement with each other and with the gun-slides.

4.2.4.4 *Elevating arc.* The contractor shall further ascertain that the elevating arc is in alinement with the axis of the cradle trunnions and has the intended tooth contact with the axis of the elevating arc and with the elevating shaft pinion on the mount at all positions of elevation.

4.2.4.5 *Air filling valve.* Each recuperator cylinder head assembly shall be mounted in a suitable test fixture and subjected to 1,000 p.s.i. pressure for leakage of the air filling valve.

4.2.4.6 *Air leaks.* Immediately after gymnastication, the reserve oil shall be drawn off and examined for a foamy or emulsified condition.

4.2.4.7 *Oil index.* After the reserve oil has been replaced (see 4.2.4.6), the oil index shall be examined for intended functioning.

4.2.4.8 *Frictions.* Manometer readings shall be taken on each recoil mechanism after assembly, after resting at least 48 hours, after gymnastication, and after proof-firing.

4.2.5 *Inspection equipment.*

4.2.5.1 *Contractor's inspection equipment.* In inspecting the supplier's inspection equipment the Government inspector will determine that the supplier has available, and utilizes correctly, gaging, measuring, and test equipment of required accuracy and precision and that the instruments are of proper type and range to make measurements of the required accuracy. The supplier will have available a set of master gages, standards, and appropriate instruments for regularly scheduled calibration of his inspection equipment. Records of such regularly scheduled calibration will be maintained by the supplier and made available for review by the Government. The calibration of gages, standards, and instruments will be periodically checked by authorized Government personnel.

4.2.5.2 *Ordnance inspection equipment.* Gages applicable to parts are referenced in the Index to List of Inspection Gages, GL B114583.

4.2.5.3 *Supplementary Quality Assurance Provision.* Unless otherwise specified, the Government shall furnish the contractor with the applicable SQAP(s) referenced in LSQAP 7148356 and LSQAP 7148374.

**4.3 Test methods and procedures.**

4.3.1 *Proof-firing test procedure.* Each recoil mechanism shall be proof-fired as specified in table II.

(a) The proof-firing for acceptance of the recoil mechanism may be combined with that of the weapon and carriage. In such combined tests, rounds in addition to those required by table II may be fired.

(b) At the conclusion of the proof-firing test, each recoil mechanism shall



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be carefully examined for leaks, breaks, or deformations. Manometer tests shall be made and pressure and frictions calculated.

(c) Velocimeter records of recoil and counter-recoil shall be taken during the proof-firing of the mechanism if such action is considered necessary by the inspector or proof-officer.

TABLE II. *Proof-firing*

| Number of Rounds | Percent of Rated Maximum Pressure | Elevation |
|---|---|---|
| 1 | Standard | 0° |
| 1 | 115% | 65° (Maximum) |
| 1 | Standard | 65° (Maximum) |
| 1 | Standard | 0° |

**4.4 Resubmission.** Rejected mechanisms shall not be resubmitted until—the mechanism has been screened and all defects removed or repaired by the contractor; the Government inspector is notified of precautionary measures instituted to prevent recurrence of failures causing rejection; and the mechanism has been marked plainly as a resubmitted mechanism.

## 5. PREPARATION FOR DELIVERY

**5.1 Recoil mechanisms.** Preparation for delivery of the recoil mechanism shall be in accordance with the applicable text of Specification MIL–M–11965.

**5.2 Repair parts.** Repair parts shall be prepared for shipment as specified on the applicable packaging data sheet referenced in the supplement to Specification MIL–W–14539.

## 6. NOTES

**6.1 Ordering data.** Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Title, number, and date of applicable drawings.

(c) List of Government furnished equipment, if any.

(d) Place of inspection.

(e) Place of tests.

(f) List of maintenance and repair parts required, with title, number, and date of applicable Supplementary Quality Assurance Provisions (see 4.2.3.3).

(g) Levels of packaging and packing for repair parts.

**6.2 Suggested contractual features.** The contractor should be furnished a block of serial numbers to be used.

**6.3 Proof-firing.** Whenever possible, the recoil mechanism should be proof-fired in combination with unproofed carriages and cannon. When this is done the following firing schedule should be used:

| *Number of Rounds* | *Type of Rounds* | *Elevation* | *Traverse* |
|---|---|---|---|
| 1 | Standard | 0° | Center |
| 1 | 115% | 65° (Maximum) | Center |
| 1 | Standard | 65° (Maximum) | * |
| 1 | Standard | 0° | * |

**These standard rounds should be fired alternately over the right and left trails.*

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